

CISCO NETWORKING LAB REPORT

Route Redistribution Mega Lab

Multi-Protocol Redistribution: OSPF • EIGRP • RIP v2
40 Routers | 78 Total Networks | Advanced Configuration

1. Lab Overview

This lab demonstrates advanced multi-protocol route redistribution across a large-scale Cisco network consisting of 40 routers organized into four distinct routing protocol domains. The objective is to achieve full network reachability among all 78 networks through proper redistribution configuration at boundary routers.

Protocol	Domain	Networks	Routers
OSPF 10	Areas 0, 1, 2, 3	43	R1 – R23
EIGRP 100	AS 100	10	R23 – R28
RIP v2	Classless	12	R28 – R34
EIGRP 50	AS 50	13	R31, R35 – R40
TOTAL NETWORKS (after redistribution)			78

2. Network Topology

The following diagram illustrates the complete network topology with all routing protocol domains, redistribution points, and inter-router connections.

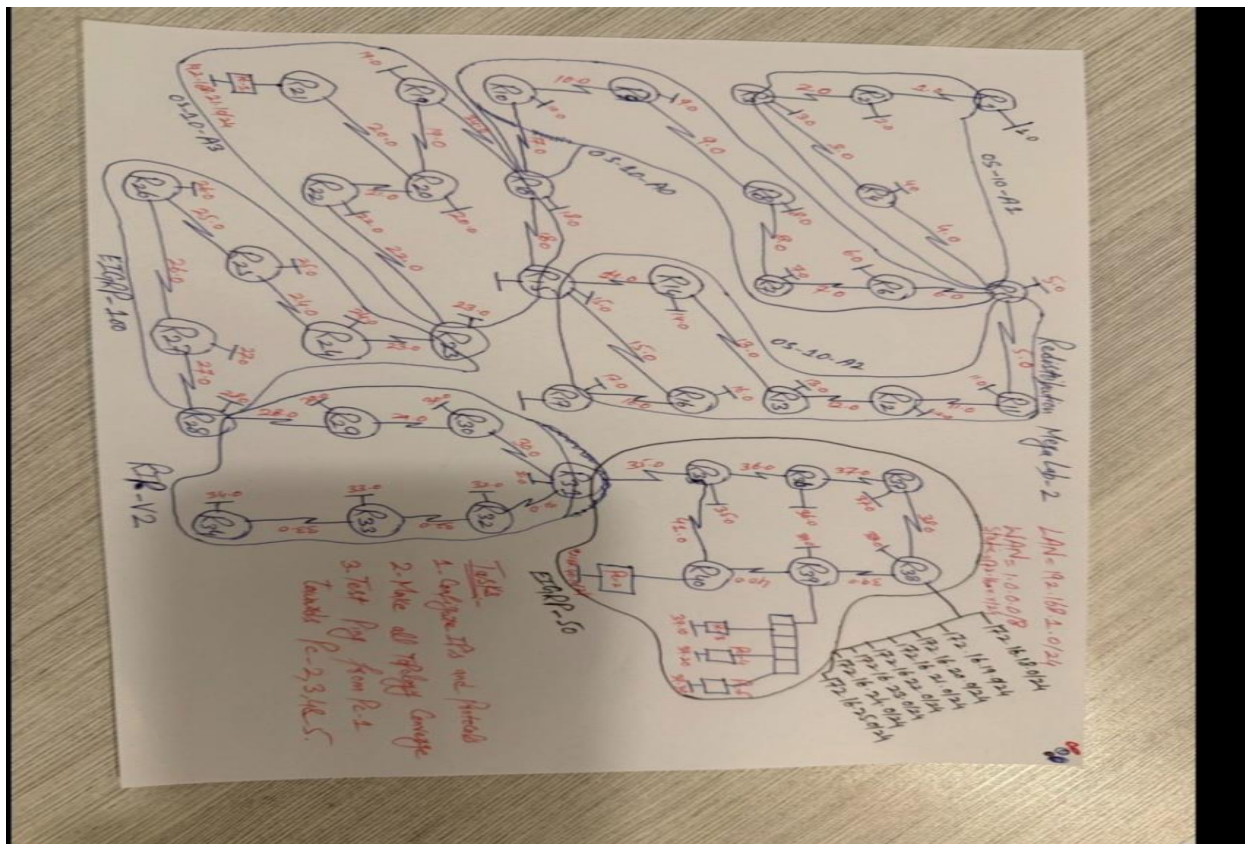


Figure 1: Complete Network Topology – 40-Router Redistribution Lab

3. IP Addressing Table

The table below lists all router interface assignments including port numbers, routing protocol associations, IP addresses, and subnet masks across the entire lab topology.

Router No	Port No	Protocol	IP Address	Subnet Mask
R1	S1/0	OSPF 10 A1	1.1.1.1	255.0.0.0
R1	E0/0	Null	192.168.1.100	255.255.255.0
R2	E0/0	OSPF 10 A1	192.168.2.100	255.255.255.0
R2	S1/0	OSPF 10 A1	1.1.1.2	255.0.0.0
R2	S1/1	OSPF 10 A1	2.1.1.1	255.0.0.0
R3	E0/0	OSPF 10 A1	192.168.3.100	255.255.255.0
R3	S1/1	OSPF 10 A1	2.1.1.2	255.0.0.0
R3	S1/0	OSPF 10 A1	3.1.1.1	255.0.0.0
R4	E0/0	OSPF 10 A1	192.168.4.100	255.255.255.0
R4	S1/0	OSPF 10 A1	3.1.1.2	255.0.0.0
R4	S1/1	OSPF 10 A1	4.1.1.1	255.0.0.0
R5	E0/0	Null	192.168.5.100	255.255.255.0
R5	S1/1	OSPF 10 A1	4.1.1.2	255.0.0.0
R5	S1/0	OSPF 10 A0	6.1.1.1	255.0.0.0
R5	S1/2	OSPF 10 A2	5.1.1.1	255.0.0.0
R6	E0/0	OSPF 10 A0	192.168.6.100	255.255.255.0
R6	S1/0	OSPF 10 A0	6.1.1.2	255.0.0.0
R6	S1/1	OSPF 10 A0	7.1.1.1	255.0.0.0
R7	E0/0	OSPF 10 A0	192.168.7.100	255.255.255.0
R7	S1/1	OSPF 10 A0	7.1.1.2	255.0.0.0
R7	S1/0	OSPF 10 A0	8.1.1.1	255.0.0.0
R8	E0/0	OSPF 10 A0	192.168.8.100	255.255.255.0
R8	S1/0	OSPF 10 A0	8.1.1.2	255.0.0.0
R8	S1/1	OSPF 10 A0	9.1.1.1	255.0.0.0
R9	E0/0	OSPF 10 A0	192.168.9.100	255.255.255.0
R9	S1/1	OSPF 10 A0	9.1.1.2	255.0.0.0
R9	S1/0	OSPF 10 A0	10.1.1.1	255.0.0.0
R10	E0/0	OSPF 10 A0	192.168.10.100	255.255.255.0
R10	S1/0	OSPF 10 A0	10.1.1.2	255.0.0.0
R10	S1/1	OSPF 10 A0	17.1.1.1	255.0.0.0

Router No	Port No	Protocol	IP Address	Subnet Mask
R11	E0/0	OSPF 10 A2	192.168.11.100	255.255.255.0
R11	S1/2	OSPF 10 A2	5.1.1.2	255.0.0.0
R11	S1/0	OSPF 10 A2	11.1.1.1	255.0.0.0
R12	E0/0	OSPF 10 A2	192.168.12.100	255.255.255.0
R12	S1/0	OSPF 10 A2	11.1.1.2	255.0.0.0
R12	S1/1	OSPF 10 A2	12.1.1.1	255.0.0.0
R13	E0/0	OSPF 10 A2	192.168.13.100	255.255.255.0
R13	S1/1	OSPF 10 A2	12.1.1.2	255.0.0.0
R13	S1/0	OSPF 10 A2	13.1.1.1	255.0.0.0
R14	E0/0	OSPF 10 A2	192.168.14.100	255.255.255.0
R14	S1/0	OSPF 10 A2	13.1.1.2	255.0.0.0
R14	S1/1	OSPF 10 A2	14.1.1.1	255.0.0.0
R15	E0/0	OSPF 10 A2	192.168.15.100	255.255.255.0
R15	S1/1	OSPF 10 A2	14.1.1.2	255.0.0.0
R15	S1/0	OSPF 10 A2	15.1.1.1	255.0.0.0
R15	S1/2	OSPF 10 A3	18.1.1.1	255.0.0.0
R16	E0/0	OSPF 10 A2	192.168.16.100	255.255.255.0
R16	S1/0	OSPF 10 A2	15.1.1.2	255.0.0.0
R16	S1/1	OSPF 10 A2	16.1.1.1	255.0.0.0
R17	E0/0	OSPF 10 A2	192.168.17.100	255.255.255.0
R17	S1/1	OSPF 10 A2	16.1.1.2	255.0.0.0
R18	E0/0	NULL	192.168.18.100	255.255.255.0
R18	S1/1	OSPF 10 A0	17.1.1.2	255.0.0.0
R18	S1/0	OSPF 10 A3	34.1.1.1	255.0.0.0
R18	S1/2	OSPF 10 A3	18.1.1.2	255.0.0.0
R19	E0/0	OSPF 10 A3	192.168.19.100	255.255.255.0
R19	S1/0	OSPF 10 A3	34.1.1.2	255.0.0.0
R19	S1/1	OSPF 10 A3	19.1.1.1	255.0.0.0
R20	E0/0	OSPF 10 A3	192.168.20.100	255.255.255.0
R20	S1/1	OSPF 10 A3	19.1.1.2	255.0.0.0
R20	S1/0	OSPF 10 A3	20.1.1.1	255.0.0.0
R20	S1/2	OSPF 10 A3	21.1.1.1	255.0.0.0
R21	E0/0	OSPF 10 A3	192.168.21.100	255.255.255.0
R21	S1/0	OSPF 10 A3	20.1.1.2	255.0.0.0
R22	E0/0	OSPF 10 A3	192.168.22.100	255.255.255.0

Router No	Port No	Protocol	IP Address	Subnet Mask
R22	S1/2	OSPF 10 A3	21.1.1.2	255.0.0.0
R22	S1/0	OSPF 10 A3	22.1.1.1	255.0.0.0
R23	E0/0	OSPF 10 A3	192.168.23.100	255.255.255.0
R23	S1/0	OSPF 10 A3	22.1.1.2	255.0.0.0
R23	S1/1	EIGRP 100	23.1.1.1	255.0.0.0
R24	E0/0	EIGRP 100	192.168.24.100	255.255.255.0
R24	S1/1	EIGRP 100	23.1.1.2	255.0.0.0
R24	S1/0	EIGRP 100	24.1.1.1	255.0.0.0
R25	E0/0	EIGRP 100	192.168.25.100	255.255.255.0
R25	S1/0	EIGRP 100	24.1.1.2	255.0.0.0
R25	S1/1	EIGRP 100	25.1.1.1	255.0.0.0
R26	E0/0	EIGRP 100	192.168.26.100	255.255.255.0
R26	S1/1	EIGRP 100	25.1.1.2	255.0.0.0
R26	S1/0	EIGRP 100	26.1.1.1	255.0.0.0
R27	E0/0	EIGRP 100	192.168.27.100	255.255.255.0
R27	S1/0	EIGRP 100	26.1.1.2	255.0.0.0
R27	S1/1	EIGRP 100	27.1.1.1	255.0.0.0
R28	E0/0	EIGRP 100	192.168.28.100	255.255.255.0
R28	S1/1	EIGRP 100	27.1.1.2	255.0.0.0
R28	S1/0	RIP V2	28.1.1.1	255.0.0.0
R29	E0/0	RIP V2	192.168.29.100	255.255.255.0
R29	S1/0	RIP V2	28.1.1.2	255.0.0.0
R29	S1/1	RIP V2	29.1.1.1	255.0.0.0
R30	E0/0	RIP V2	192.168.30.100	255.255.255.0
R30	S1/1	RIP V2	29.1.1.2	255.0.0.0
R30	S1/0	RIP V2	30.1.1.1	255.0.0.0
R31	E0/0	RIP V2	192.168.31.100	255.255.255.0
R31	S1/0	RIP V2	30.1.1.2	255.0.0.0
R31	S1/1	RIP V2	31.1.1.1	255.0.0.0
R31	S1/2	EIGRP 50	35.1.1.1	255.0.0.0
R32	E0/0	RIP V2	192.168.32.100	255.255.255.0
R32	S1/1	RIP V2	31.1.1.2	255.0.0.0
R32	S1/0	RIP V2	32.1.1.1	255.0.0.0
R33	E0/0	RIP V2	192.168.33.100	255.255.255.0
R33	S1/0	RIP V2	32.1.1.2	255.0.0.0

Router No	Port No	Protocol	IP Address	Subnet Mask
R33	S1/1	RIP V2	33.1.1.1	255.0.0.0
R34	E0/0	RIP V2	192.168.34.100	255.255.255.0
R34	S1/1	RIP V2	33.1.1.2	255.0.0.0
R35	E0/0	EIGRP 50	192.168.35.100	255.255.255.0
R35	S1/2	EIGRP 50	35.1.1.2	255.0.0.0
R35	S1/1	EIGRP 50	41.1.1.2	255.0.0.0
R35	S1/0	EIGRP 50	36.1.1.1	255.0.0.0
R36	E0/0	EIGRP 50	192.168.36.100	255.255.255.0
R36	S1/0	EIGRP 50	36.1.1.2	255.0.0.0
R36	S1/1	EIGRP 50	37.1.1.1	255.0.0.0
R37	E0/0	EIGRP 50	192.168.37.100	255.255.255.0
R37	S1/1	EIGRP 50	37.1.1.2	255.0.0.0
R37	S1/0	EIGRP 50	38.1.1.1	255.0.0.0
R38	E0/0	EIGRP 50	192.168.38.100	255.255.255.0
R38	S1/0	EIGRP 50	38.1.1.2	255.0.0.0
R38	S1/1	EIGRP 50	39.1.1.1	255.0.0.0
R39	E0/0	EIGRP 50	192.168.39.100	255.255.255.0
R39	S1/1	EIGRP 50	39.1.1.2	255.0.0.0
R39	S1/0	EIGRP 50	40.1.1.1	255.0.0.0
R40	E0/0	EIGRP 50	192.168.40.100	255.255.255.0
R40	S1/0	EIGRP 50	40.1.1.2	255.0.0.0
R40	S1/1	EIGRP 50	41.1.1.1	255.0.0.0

4. End-Device (PC) Addressing Table

The following table contains the IP addressing details for all PCs configured in this lab. PCs are used for end-to-end connectivity verification through pings across different routing domains.

PC No	IP Address	Subnet Mask	Default Gateway
PC 1	192.168.21.10	255.255.255.0	192.168.21.100
PC 2	192.168.40.10	255.255.255.0	192.168.40.100
PC 3	192.168.39.10	255.255.255.0	192.168.39.100
PC 4	192.168.39.20	255.255.255.0	192.168.39.100
PC 5	192.168.39.30	255.255.255.0	192.168.39.100

5. Static IP Assignments – Router R38

Router R38 is configured with the following static loopback/null routes to simulate additional networks for redistribution into EIGRP AS 50. These networks are redistributed using the redistribute static command.

Network Address	Subnet Mask
172.16.18.100	255.255.255.0 (/24)
172.16.19.100	255.255.255.0 (/24)
172.16.20.100	255.255.255.0 (/24)
172.16.21.100	255.255.255.0 (/24)
172.16.22.100	255.255.255.0 (/24)
172.16.23.100	255.255.255.0 (/24)
172.16.24.100	255.255.255.0 (/24)
172.16.25.100	255.255.255.0 (/24)

6. Configuration Commands

The following IOS commands are used to configure interfaces, routing protocols, and redistribution across all boundary routers. Commands are executed in privileged EXEC and global configuration modes.

6.1 Interface & Basic Setup

```
en
config t
int e0/0 (or s1/0, s1/1, s1/2)
ip addr <IP_ADDRESS> <SUBNET_MASK>
no sh
wr
```

6.2 Static Route for Redistribution (R38)

```
ip route 172.16.25.100 255.255.255.0 Null 0
redistribute static
redistribute connected
```

6.3 OSPF 10 Redistribution

```
router ospf 10
network 192.168.x.x 0.0.0.0 area 0
redistribute eigrp 100 subnets metric 1
redistribute rip subnets metric 1
```

6.4 EIGRP 100 Redistribution

```
router eigrp 100
network 192.168.x.x 0.0.0.0
redistribute ospf 10 metric 1 1 1 1 1
redistribute rip metric 1 1 1 1 1
```

6.5 RIP v2 Redistribution

```
router rip
version 2
network 192.168.x.0
redistribute ospf 10 metric 1
redistribute eigrp 100 metric 1
```

6.6 Verification Commands

```
show ip route summary
show run
do wr
```


7. Lab Screenshots

7.1 Lab Physical Setup

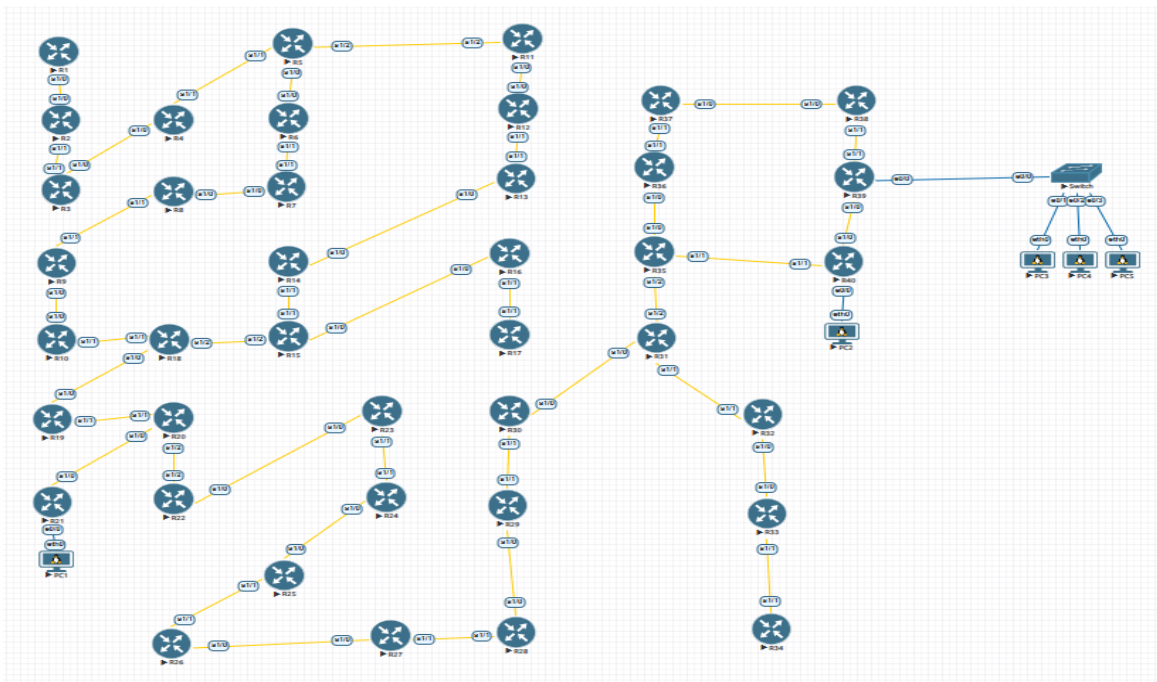


Figure 2: Physical Lab Setup

7.2 Routing Table Verification

```

R40#show ip route sum
IP routing table name is default (0x0)
IP routing table maximum-paths is 32
Route Source      Networks      Subnets      Replicates      Overhead      Mem
ory (bytes)
connected         0             6             0               408           108
0
static            0             0             0               0             0
application       0             0             0               0             0
eigrp 50          75           0             0              6992          135
00
internal         3
0
Total            78           6             0              7400          188
80

```

Figure 4: Routing Table show ip route output

7.4 Additional Verification Screenshots

```

VPCS> ping 192.168.39.10

84 bytes from 192.168.39.10 icmp_seq=1 ttl=49 time=158.244 ms
84 bytes from 192.168.39.10 icmp_seq=2 ttl=49 time=150.712 ms
84 bytes from 192.168.39.10 icmp_seq=3 ttl=49 time=144.828 ms
84 bytes from 192.168.39.10 icmp_seq=4 ttl=49 time=144.777 ms
84 bytes from 192.168.39.10 icmp_seq=5 ttl=49 time=149.103 ms

VPCS> ping 192.168.39.20

84 bytes from 192.168.39.20 icmp_seq=1 ttl=49 time=147.405 ms
84 bytes from 192.168.39.20 icmp_seq=2 ttl=49 time=144.965 ms
84 bytes from 192.168.39.20 icmp_seq=3 ttl=49 time=150.907 ms
84 bytes from 192.168.39.20 icmp_seq=4 ttl=49 time=140.305 ms
84 bytes from 192.168.39.20 icmp_seq=5 ttl=49 time=151.758 ms

```

Figure 5: Route Summary Verification

```
VPCS> ping 192.168.39.30
```

```
84 bytes from 192.168.39.30 icmp_seq=1 ttl=49 time=159.193 ms
84 bytes from 192.168.39.30 icmp_seq=2 ttl=49 time=138.873 ms
84 bytes from 192.168.39.30 icmp_seq=3 ttl=49 time=143.007 ms
84 bytes from 192.168.39.30 icmp_seq=4 ttl=49 time=148.766 ms
84 bytes from 192.168.39.30 icmp_seq=5 ttl=49 time=149.677 ms
```

Figure 6: PC Ping Test End-to-End Connectivity Verification

8. Connectivity Verification

After completing the redistribution configuration on all boundary routers, end-to-end connectivity was verified by issuing ICMP ping tests between PCs located in different routing protocol domains.

8.1 Ping Test Matrix

Source	Source IP	Destination IP	Dest	Result
PC 1	192.168.21.10	192.168.40.10	PC 2	Success
PC 1	192.168.21.10	192.168.39.10	PC 3	Success
PC 1	192.168.21.10	192.168.39.20	PC 4	Success
PC 1	192.168.21.10	192.168.39.30	PC 5	Success

9. Lab Conclusion

This lab successfully demonstrates the configuration and verification of route redistribution across four routing protocol domains on a 40-router Cisco network. All 78 networks were reachable after applying the redistribution configuration at boundary routers R5, R23, R28, and R31. The ping tests from PC 1 to PC 2, 3, 4, and 5 confirmed full end-to-end connectivity across OSPF, EIGRP, and RIP domains.

Key Takeaways:

- ▶ Redistribution requires metric translation between incompatible protocols.
- ▶ Boundary routers (R5, R23, R28, R31) are configured on both protocol domains.
- ▶ The subnets keyword is required in OSPF when redistributing classful protocols.
- ▶ Static routes on R38 were redistributed into EIGRP 50 using redistributed static.